



Mushaba: Your Companion for Hajj & Umrah

Tagline: Never lose your way — stay connected, stay together.

Overview

Every year, millions of people travel for Hajj and Umrah. During this spiritual journey, one of the biggest problems they face is moving through large, crowded places and staying close to their family or group. With so many similar-looking gates, buildings, and huge crowds, it's easy to get lost or separated, which can cause worry and stress.

Mushaba is a mobile app made to solve this problem. It helps pilgrims share their live



location, stay connected with their loved ones, and easily find their way back to their group. This makes the journey safer, calmer, and more focused on worship.

The Problem Mushaba Solves

During Hajj or Umrah, pilgrims frequently lose contact with their companions due to:

- **Massive Crowds:** Physical separation is almost inevitable during Tawaf, Sa'i, or moving between Mina and Arafat.
- **Structural Complexity:** Landmarks look identical; GPS often fails to distinguish between the floor levels you are on.
- **Connectivity Blackouts:** High-density areas often face cellular data failures, and not all family members have SIM cards. Without mobile or Wi-Fi networks, it becomes difficult to reconnect with the group.

- **Vulnerable Demographics:** Children and elderly pilgrims may not know how to use a smartphone or communicate their location to strangers.
- **Language Barriers:** Communicating with local authorities or other pilgrims is difficult for international travelers.

The Mushaba Solution

Mushaba transforms the challenging experience of potential separation into a safe, traceable, and manageable process. The mobile application allows pilgrims to:

- **Create or Join Private Groups:** By just scanning a QR Code, you can manage users Just like a familiar chat group, users can form private circles with their family or tour group.
- **Share Real-time GPS Location:** See the precise location of every group member on a shared Map interface along with the floor number.
- **Access Floor Indication:** Know not just where someone is, but also which floor they are on (e.g., basement, 1st, 2nd, 3rd floor).
- **AI Voice Translator:** Automatically detects spoken language in real time and translates it into the listener's native language, while also converting the listener's response back into the speaker's original language for seamless two-way communication (**Voice to Voice**).
- **Main Screen Shortcut Access:** Communicate quickly and clearly with members, easily share voice notes, images, and pinned locations, enable AI Lost Mode, and monitor others' real-time locations along with their floor numbers.
- **Share Pin Location:** Mark important spots or your standing point for others to see and send real-time photos of your exact surroundings (e.g., a specific gate, shop, restaurant or sign) to aid recognition.
- **Utilize "AI Lost Mode" Assistance:** If separated, an AI guide helps navigate you back to your group, a designated meeting point, bus station, or hotel and send SOS alerts to the group members.
- **Engage in In-App Chat:** Communicate individually or with the entire group through text messages and share real-time messages, voice notes, photos, videos, pin locations, and send SOS alerts.
- **Customizable QR Bands for Kids & Elders:** Generate QR bands for non-smartphone users. When scanned, they display the pilgrim's name, group, one-tap caretaker call, and hotel or bus GPS location—helping lost members reconnect with their group quickly.



Mushaba brings clarity to confusion, connection to isolation, and peace of mind to what can be a crowded environment, allowing pilgrims to fully focus on their worship and spiritual reflection.

Key Features – In Detail



I. Smart Group Creation & Management

Private Groups via QR Code

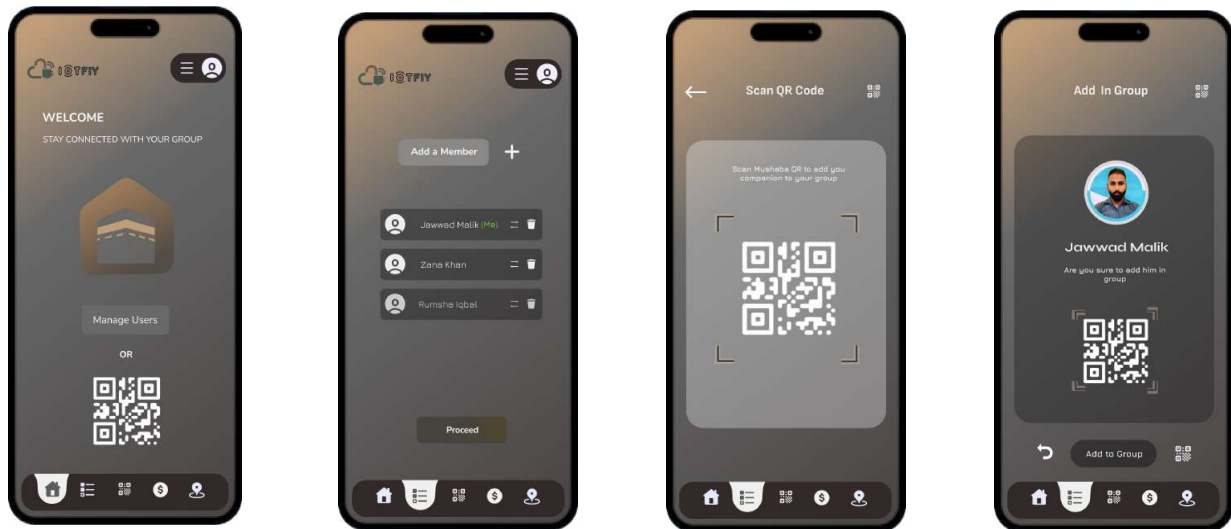
Users can instantly **create or join private groups** by scanning a QR code—similar to joining a familiar chat group. These groups can represent families, tour batches, or caretaking units.

Admin-Controlled Groups

Each group is managed by an admin who can:

- Add or remove members
- Share group-wide updates
- Assign or transfer leadership

This structure is ideal for tour operators and group leaders.



II. Comprehensive Location Tracking & Navigation

Live Group Location Sharing

View all group members in **real time on a shared map**, with each user clearly labeled by name—making regrouping fast and effortless.

Floor-Level Location Awareness: Mushaba shows **exact floor information** (Basement, Ground, 1st, 2nd, 3rd floor, etc.), essential for complex multi-level areas like mosques, terminals, and malls.



Distance & Direction Indicators

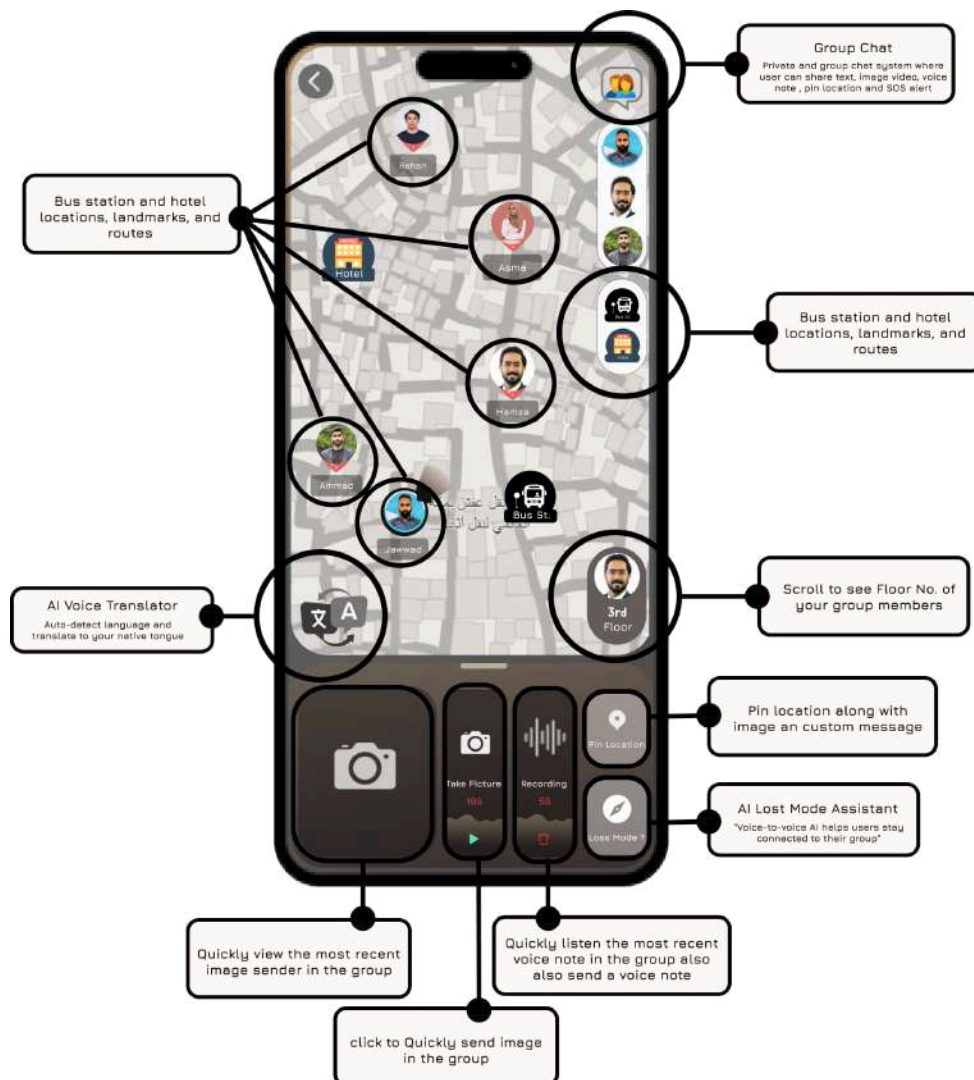
See **how far** and **in which direction** each member is located, reducing confusion in dense crowds.

One-Tap Hotel & Bus Station Navigation

Pinned **Hotel and Bus Station locations** are always accessible.

Users can tap any pin to:

- Instantly view the route
- Get guided directions
- Reach destinations quickly without asking for help



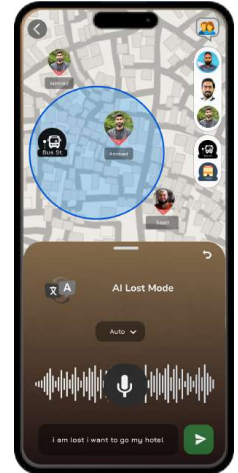
III. AI-Powered Lost Assistance

AI “Lost Mode”

When someone gets separated, activating **AI Lost Mode**:

- Guides the user back to their group, hotel, or bus station
- Calculates the most efficient route based on surroundings
- Sends automatic alerts to group members

This ensures fast recovery with minimal panic.



IV. AI Voice & Language Translator

Real-Time Voice-to-Voice Translation

Mushaba automatically detects spoken language and translates it **live** into the listener’s native language—then translates the response back into the original language for smooth two-way communication.

Auto-Translated Chat Messages

All text messages are **instantly translated** into the recipient’s preferred language (Arabic, Urdu, English etc.), removing communication barriers.



V. Offline & Internet-Free Connectivity

Mesh Networking (No Internet Required)

Even without SIM, Wi-Fi, or mobile data, Mushaba enables:

- Location sharing
- Floor-level tracking
- Group visibility

Bluetooth & Wi-Fi Direct Communication



The app creates a **secure peer-to-peer mesh network**.

If any device regains internet access, it automatically syncs updates for the entire group—without overloading a single device.

VII. Customizable QR Safety Bands (Kids & Elders)

Digital-to-Physical Safety

For children and elders without smartphones, admins can generate **custom QR bands** linked to their group.

Public Scan Recovery (No App Required)

When scanned by anyone, the QR code displays:

- Pilgrim's name & group
- One-tap call to caretaker
- Hotel or bus GPS location (clickable for routing)
- Pre-translated recovery instructions (Arabic & English)

This enables **instant assistance from the public**.



VI. Communication & Coordination Suite

Unified In-App Chat

Communicate through:

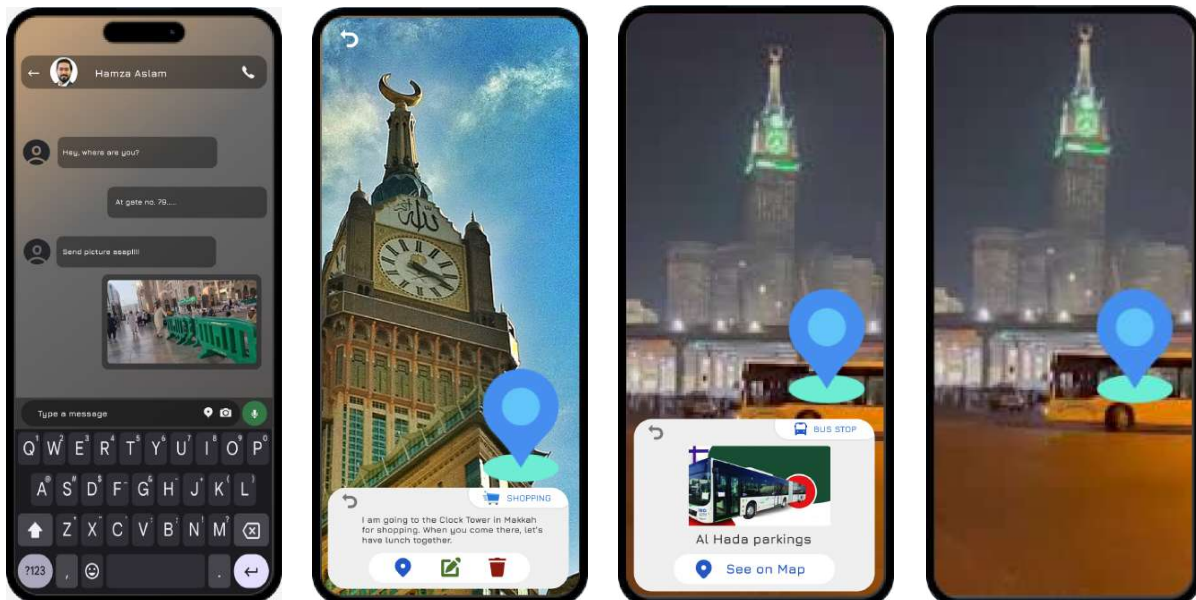
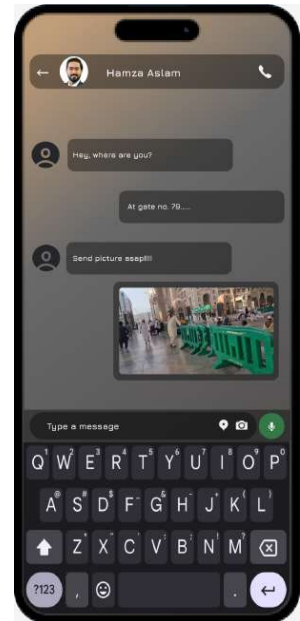
- Text messages
- Voice notes
- Images & videos
- Location pins
- SOS alerts

Visual Landmark Sharing

Users can take and send **real-time photos** of surroundings (gates, signs, shops, landmarks) to help others identify locations faster.

Pin Location Sharing

Drop meeting points or important locations on the map so everyone can follow the same reference point.



VIII. Privacy & Security

Private & Secure Groups

All location data and communication remain **strictly within private groups**.

Controlled Access

Groups can be protected via:

- Admin-only invites
- Optional passwords

No external user can access group data.

IX. The Spiritual Suite

1. Interactive Animated Guide & Contextual Dua Assistant This feature ensures that every pilgrim, regardless of their prior knowledge or background, can perform their rituals with confidence and clarity.

- **Multi-Lingual Animated Walkthroughs:** Step-by-step visual guides utilizing modern, minimalist animations to explain each stage of Hajj and Umrah clearly.
- **Universal Language Support:** Audio and text instructions are fully localized, allowing users to understand the precise requirements of their journey in their native language.
- **Context-Aware Dua Helper:** The app automatically provides the exact supplications (Duas) required for the specific ritual being performed, presented in a distraction-free brown and dark grey interface to maintain spiritual focus.

2. Automated Ritual Tracking & Assurance By leveraging on-device sensors and location data, the application acts as a smart digital guide to ensure all rituals are completed smoothly and strictly according to the teachings of Prophet Muhammad (PBUH).

- **Smart Lap Counter:** Automatically tracks the user's movement to accurately count the 7 circuits around the Kaaba during Tawaf and the 7 trips between Safa and Marwah during Sa'i.
- **Step-by-Step Umrah Tracker:** Automatically monitors and logs the completion of the 7 key Umrah milestones: Ihram (Niyyah/Talbiyah at Miqat), Entering Masjid al-Haram (with Dua), Tawaf, Praying 2-rak'ah at Maqam Ibrahim, Drinking Zamzam water, Sa'i, and Halq/Taqsir.

- **Comprehensive Hajj Journey Logging:** Provides a real-time checklist and automated tracking for all critical Hajj stages, including entering Ihram, staying in Mina, standing at Arafat, staying in Muzdalifah, the stoning of Jamarat, animal sacrifice, shaving hair, Tawaf al-Ifadah, Sa'i, and the Farewell Tawaf.
- **Sunnah Compliance Alerts:** Sends gentle, timely reminders for upcoming steps to prevent missed rituals and provide peace of mind throughout the journey.

Real-World Use Cases

- **Family Groups During Umrah or Hajj**

Fatima becomes separated from her family after prayer due to heavy crowd movement. Using Mushaba, she instantly locates her brother **230 meters away near Gate 14**, complete with floor indication. Her brother shares a live photo of the surrounding landmarks, helping their elderly father—who is less tech-savvy—follow clear visual directions and reunite safely.

- **Group Tour Operators & Hajj Agencies**

A Hajj agency manages a group of **50 pilgrims** through Mushaba, with the guide acting as the group admin. The guide monitors every pilgrim's **real-time location and floor level** on a single dashboard, allowing immediate response if anyone strays. Lost-person incidents drop drastically, improving **operational efficiency, safety, and client trust**.

- **Elderly Pilgrims & First-Time Travelers**

Mohammed, 68, visiting from Indonesia, becomes disoriented inside a multi-level area. He activates **AI Lost Mode**, which alerts his group and shares his exact location and surroundings. A nearby group member receives directions and reaches him within minutes—preventing panic and ensuring a calm recovery.

- **Families with Children**

During Tawaf, a child briefly moves out of sight in the dense crowd. Instead of panic, the parents open Mushaba and instantly see the child's **precise location**

and floor level on the map. They follow the guided route and reunite within moments, turning a potentially dangerous situation into a controlled and peaceful outcome.

- **Children or Elders Wearing QR Safety Bands**

A young child wearing a Mushaba QR band is found alone near a mosque entrance. A passerby scans the QR code (no app required) and instantly sees the child's **name, group details, and a one-tap call option** for the caretaker. The screen also shows the group's **hotel and bus location with clickable navigation**, enabling immediate assistance.

- **Language Barrier Situations**

A pilgrim from Turkey needs urgent help but cannot communicate with nearby volunteers. Using Mushaba's **AI voice translator**, both parties speak in their own languages while the app handles real-time translation. Clear communication prevents confusion and speeds up assistance.

- **Internet Blackout or Network Congestion**

During peak Hajj hours, mobile networks become unreliable. Despite no internet access, Mushaba continues to function through **Bluetooth and Wi-Fi Direct mesh networking**, allowing pilgrims to see nearby group members' locations and floors—ensuring safety even in zero-connectivity zones.

- **Emergency Evacuation or Crowd Redirection**

In the event of sudden crowd redirection, a group admin pins a **safe meeting point** and the **bus station location** on the map. Members simply tap the pin to view the fastest route and regroup efficiently, avoiding misinformation and chaos.

Why Mushaba Matters

"Mushaba is more than just a navigation app; it is a lifeline in crowded spiritual environments. By ensuring pilgrims are always one tap away from their loved ones, it allows them to focus entirely on their faith. Designed for peace of mind, Mushaba transforms the chaos of a crowd into a sense of confidence, safety, and connection."

4. Business Model & Market Strategy

Mushaba follows a **multi-channel go-to-market strategy** designed to maximize pilgrim safety while ensuring scalable and sustainable revenue. By serving governments, travel agencies, and individual families, Mushaba positions itself as a **mission-critical safety platform**, not just a mobile app.

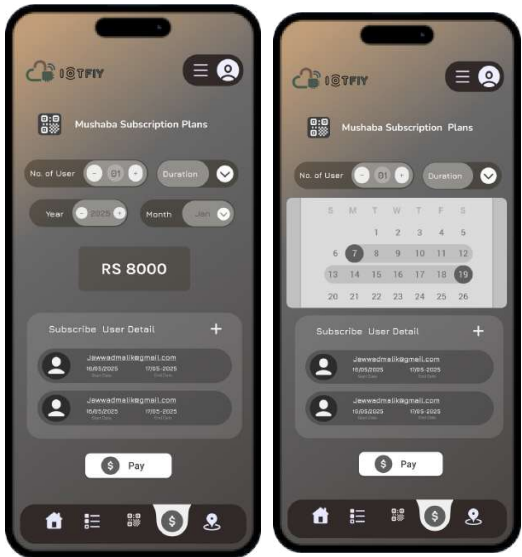
4.1 Target Market Segments

B2G – Government & Public Authorities

Mushaba partners with **Hajj and Umrah Ministries** and religious authorities to establish the platform as a **standardized safety layer** for registered pilgrims.

This enables centralized monitoring, faster recovery of lost pilgrims, and improved national safety compliance.

Value: Reduced emergency incidents, improved crowd safety, and enhanced pilgrim experience.



B2B – Travel Agencies & Tour Operators (SaaS Model)

Category	Feature	Freemium Plan	Premium Plan (B2C)
Core	User Capacity	Up to 3 users	Unlimited users

	Real-Time GPS Tracking	Basic map view	Advanced map + admin view
Communication	Text, Voice, Media	Yes	Yes
Connectivity	Offline Mesh Networking	✗	✓ (Bluetooth + Wi-Fi Direct)
Visual Tools	Landmark Photo Sharing	✗	✓ (Photo Pings)
AI Features	AI Voice Translation	✗	✓ (Two-Way, Real-Time)
	Smart “Lost Mode”	✗	✓ (AI-Guided Navigation)
Safety Tech	Floor-Level Detection	✗	✓ (3D / Multi-Level)
Physical Safety	QR Safety Bands	✗	✓ (Digital-to-Physical Recovery)
Management	Group Admin Controls	✗	✓ (Add / Remove / Broadcast)

Travel agencies receive access to a **Manager Dashboard** that allows guides to monitor and manage **50+ pilgrims per group** in real time.

This dramatically reduces operational stress, lost-person incidents, and manual coordination efforts.

Value: Operational efficiency, risk reduction, and higher client satisfaction.

B2C – Direct-to-Consumer (Families & Individuals)

Families can directly download Mushaba from the **App Store / Play Store** to protect loved ones—especially children and elders—through premium safety features.

Value: Affordable, on-demand peace of mind.

4.2 Pricing & Subscription Tiers

Subscription Plans Overview

Pricing Structure

Duration	Freemium	Premium
15-Day Access	Free	PKR 1,499 / user
40-Day Access	Free	PKR 2,999 / user

Strategic Advantage

- **Freemium entry** drives mass adoption and trust
 - **Premium safety upgrades** unlock critical features when risk is highest
 - **Government & agency partnerships** enable large-scale deployment
 - **Digital + physical safety integration** (QR bands) creates strong differentiation
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5. Current Development Stage: Deployment & Scaling

Mushaba has successfully progressed from concept to a **market-ready safety platform**. The product has completed its **MVP**, followed by rigorous **Alpha and Beta testing**, validating performance in **high-density, multi-level pilgrimage environments**.

Current Status:

Live Deployment & Market Expansion Phase

Key Milestones Achieved

- **Operational Offline Mesh Networking** enabling reliable location sharing in zero-connectivity zones
- **AI-powered real-time translation engine** supporting major pilgrim languages (Arabic, Urdu, Bahasa, English, and more)
- **Accurate floor-level detection** optimized for complex Tawaf and mosque structures

Immediate Growth Roadmap

- Finalizing **B2B partnerships with Tier-1 Umrah travel agencies**
- Strategic engagements and pilots with **regional Hajj and Umrah Ministries**
- Large-scale rollout preparation for the **2026 Hajj season**

6. Future Scope: Beyond Safety to Complete Immersion

1. Smart IoT Wearables

Physical smart bands linked to the app for **real-time location tracking** and **dynamic QR functionality**, ensuring instant assistance for pilgrims without smartphones.

2. Integrated Travel Marketplace

A **one-stop platform** for booking hotels, flights, and Hajj & Umrah packages, streamlining the entire pilgrimage travel experience.

3. AI Heritage Lens

Picture-based AI tool that **narrates the history of landmarks** captured by the camera, enhancing learning and cultural appreciation during the journey.

4. Smart Ziyarat Discovery

Interactive map alerts guide pilgrims to **nearby holy visitation sites**, offering context, timings, and optimal routes.

5. The Spiritual Companion

An AI-powered spiritual assistant providing **Umrah guidance, Dua repository, and Digital Tasbih**, creating a holistic and immersive religious experience.

7. Company & Presenter Information

Product Ownership

Mushaba is a flagship safety and navigation product developed by **IOTFIY Solutions**.

About IOTFIY Solutions

IOTFIY Solutions is a **product- and services-based technology company** specializing in **IoT systems, Artificial Intelligence, and mobile application development**.

The company focuses on **innovation-driven solutions and product incubation**, building scalable technologies for real-world, high-impact use cases across multiple product lines.

Leadership

Founder & CEO: Muhammad Jawwad Malik

Vision Statement:

“Our mission is to make large-scale pilgrimages safer, smarter, and stress-free through intelligent technology.”

Team & Advisory Support

- **Core Team Size:** 18–25 professionals (Engineering, AI, Mobile, IoT, Product, Operations)
 - **Advisor:**
Farhan Zaheer – COO, *Nucleus Distribution Pvt. Ltd*
(Operational scaling, enterprise execution, and market strategy)
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Company Registration & Compliance

- Registered Partnership Company in **Pakistan**
 - Officially registered with:
 - **PSEB** (Pakistan Software Export Board)
 - **P@SHA** (Pakistan Software Houses Association)
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Official Links & Contact

- **Mushaba Product Page:**
<https://iotfiysolutions.com/products/mushaba.html>
 - **Company Website:**
<https://iotfiysolutions.com>
 - **LinkedIn:**
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-

8. Geographic Focus & Expansion Strategy

Initial Target Markets

Mushaba is designed for regions with **high outbound pilgrimage volume** and strong demand for safety solutions:

- Pakistan
- India
- Bangladesh
- Indonesia
- Malaysia
- Turkey

These markets represent millions of pilgrims annually and align strongly with Mushaba's safety-first use cases.

Expansion Roadmap

- **Phase 1:** South & Southeast Asia (agency partnerships + B2C adoption)
- **Phase 2:** Middle East collaboration with Umrah and Hajj authorities
- **Phase 3:** Government-level integrations and large-scale mandates

This phased approach ensures **operational readiness, regulatory alignment, and scalable deployment.**



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